

How to use this template. This is the **skeleton** of the final architecture document. Each section carries a short *grey guidance note* describing what it must contain, and *[bracketed slots]* to replace with the project's own content. Replace every slot, insert the diagrams where marked, and **delete the guidance notes** before submission. The companion *guide* explains the reasoning behind each section; the worked *example* shows a complete version at the grade-5 standard. Quality of reasoning matters more than length: a small, well-justified, well-evaluated system scores higher than a large one that cannot be explained.

Project Title

D7065E, Architecture Document

Team name: First member & Second member

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Checklist before submission**5****1. Summary**

One short paragraph: what the system does and the single guiding principle that threads through the design. Then the summary table, and a brief note of what changed since the Week 2 proposal and why.

[Write the one-paragraph summary here.]

Field	Value
Project title	
Team members	
Use case	
Repository	
Version / date	

Changes since the proposal.

[List the design changes made since the proposal, and the evidence that forced each one. If nothing changed, state that and say why.]

2. Use case and building context

Describe the building and what BuildSim provides (rooms, sensors, actuators, 3D view). State the use case and the tension or objective that makes the autonomous decision non-trivial. BuildSim holds the shared building state; this project provides the physics, the intelligence, and the control around it.

[Describe the use case and the building context here.]

3. Requirements

List functional and non-functional requirements with stable IDs, so each can be traced to a test in the test plan and to results in the evaluation. Keep them verifiable (avoid “fast” or “reliable” without a number).

ID	Requirement	How it is verified
FR-1		
FR-2		
NFR-1		
NFR-2		

4. Architecture**4.1 Context diagram (C4 Level 1)**

The system as a single box, with the people and external systems it interacts with (including BuildSim). One paragraph of narrative explaining the boundaries and the main interactions.

[Insert: Context diagram (C4 Level 1)]

[Narrative for the context diagram.]

4.2 Container diagram (C4 Level 2)

Open the system into its separately-running parts (sensor services, autonomous service, actuators, storage, dashboard) and how they communicate. Explain why the system is split this way.

[Insert: Container diagram (C4 Level 2)]

[Narrative for the container diagram.]

4.3 Component diagram (C4 Level 3)

Zoom into one container that is worth explaining (often the autonomous service) and show its internal components and their responsibilities.

[Insert: Component diagram (C4 Level 3)]

[Narrative for the component diagram.]

4.4 Design decisions and alternatives

The core of the architecture grade. For each significant decision, state the choice, the alternatives considered, and the evidence or reasoning for choosing it over them. A decision with no rejected alternative is not yet justified.

Decision	Alternatives considered	Why this one

5. Interfaces and data contracts

The contracts between components: API endpoints and/or message topics, with payload schemas and an example message. Precise enough that a component could be re-implemented from this section alone.

[Define the interfaces and data contracts here (endpoints/topics, schemas, example payloads).]

6. Simulating the sensor values (the physical model)

How sensor values are produced. State the per-room physical model (temperature heat balance, CO₂ from occupancy and ventilation) and, critically, how actuator commands feed back so the control loop closes. Include the occupancy model (for example, a head-count per room sampled conditional on room type and time of day). Give the governing equations.

[Describe the physical and occupancy models, with equations, here.]

7. Autonomous services and data pipeline

7.1 The autonomous service

The decision logic: inputs, the algorithm or rules, and outputs. Rule-based control is fully acceptable if justified; if a learned model is used, explain training, data, and the fallback when data is thin.

[Describe the autonomous service here.]

7.2 The data pipeline

How readings are collected, stored, and served to the autonomous service and the dashboard. State what is retained and for how long.

[Describe the data pipeline here.]

8. Behaviour

8.1 Key scenario (dynamic diagram)

Walk one important scenario end-to-end (for example, a room filling up) as a C4 dynamic / sequence diagram, with a short narrative of the message order.

[Insert: Dynamic / sequence diagram for the key scenario]

[Narrative for the key scenario.]

8.2 Process lifecycle (state diagram)

The lifecycle of a key process or zone (for example, setback → active → fault), as a state diagram with the transitions.

[Insert: State diagram for the process lifecycle]

[Narrative for the state diagram.]

9. Deployment and component view

Where each process runs (edge nodes, a server, BuildSim) and how they are connected. Note where computation belongs and why.

[Insert: Deployment diagram]

[Narrative for the deployment view.]

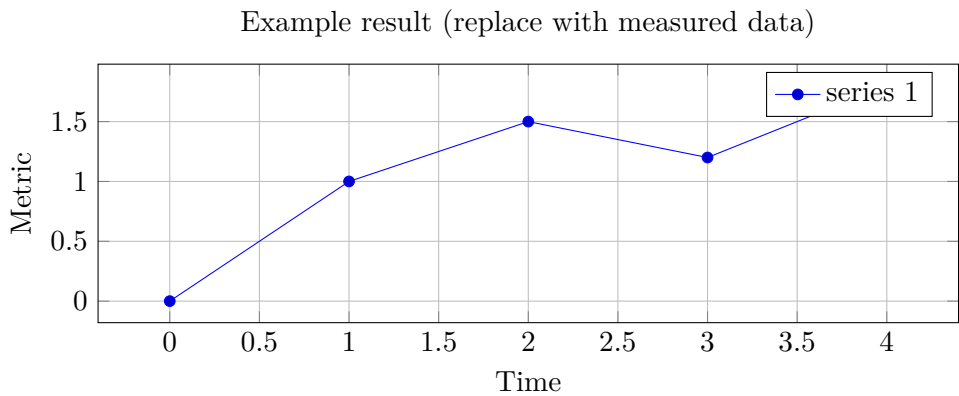
10. Test plan

The tests that verify the requirements, including stress and failure tests (a sensor that lies, a process that dies, a flood of readings). Each row should trace back to a requirement ID from Section 3.

ID	Test	Requirement	Expected result
T-1			
T-2			
T-3 (stress)			

11. Evaluation and results

The evidence that the system works, measured, not asserted. Charts and tables of results, compared against the requirements. Include an honest reading of where the numbers are weak. The plot below is a pgfplots placeholder, replace the coordinates with real data.



[Present and interpret the results here.]

12. Dashboard

What the dashboard shows and how it overlays state and decisions on the BuildSim 3D view. Include a screenshot.

[Insert: Dashboard screenshot]

[Describe the dashboard here.]

13. Risks and critical reflection

An honest account of weaknesses, trade-offs, and what would be done with more time. Critical thinking is graded here: naming a real limitation scores better than claiming there are none.

[Reflect critically on the design here: limitations, trade-offs, and next steps.]

Checklist before submission

- All *[bracketed slots]* replaced with real content.
- All grey guidance notes deleted.
- All five diagrams inserted and referenced in the text (context, container, component, dynamic, deployment).
- Every requirement in Section 3 traced to a test in Section 10.
- Each design decision in Section 4.4 names a rejected alternative.
- Evaluation backed by measured data, not assertions.
- The document matches what was actually built.